IROHUB

DATA SCIENCE

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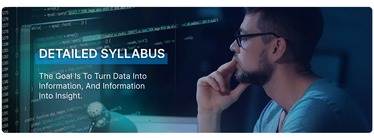


DETAILED SYLLABUS

The Goal Is To Turn Data Into Information, And Information Into Insight.

1. Introduction to Python Programming

i. Overview about Python
ii. Operators
iii. Conditional Statements
iv. Looping Statements
v. Data Types in Python
vi. Indexing and Slicing
vii. Custom Functions and Modules
viii. Object oriented Programming Concepts
ix. Exception Handling
x. Logging
xi. Introduction to GIT and GITHUB
xii. Assessment - 1

2. Numerical Programming in Python

i. Fundamentals on Data Structures and Algorithms
ii. Efficient String Operations
iii. Mastering Recursion
iv. NumPy
v. Pandas
vi. Mastering Data Wrangling
vii. Advanced Data Wrangling Concepts
vii. Data Wrangling with Code Optimization
vii. Data Management Libraries
vii. Web Scraping using Python
viii. Data Visualization - Matplotlib and Seaborn
Exploratory Data Analysis With Python and Pandas
xii. Assessment - 2

3. Relational Databases

i. SQL basics for Data Analysis
ii. Fundamentals on SQL Querying
iii. Dealing with multiple Tables
iv. Advanced SQL joins
v. Type Casting and Math Functions
vi. DateTime and String Functions
vii. Data Exploration with SQL and Python
vii. Database Management Patterns and Schema Design

4. Advanced Data Visualization

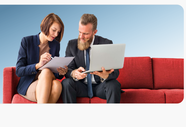
i. Fundamentsals of SQL
ii. Data Exploration in Excel with in-built functions
iii. Excel Dashboarding Concepts
iv. Advanced Dashboarding concepts with Tableau
v. Getting started with Power BI
vi. Deep Dashboarding principles
xii. Assessment - 4

5. Applied Statistics

i. Maths Refresher
ii. Foundational Mathematics for Data Science
iii. Advanced Computations
iv. Probability theory
v. Discrete Probability Distributions
vi. Continuous Probability Distributions
vii. Joint Distribution Concepts
viii. Inferential Statistics
ix. Sampling and Statistical Inference
x. Hypothesis testing
xi. Assessment - 2

6. Applied Statistics

i. Getting started with ML
ii. Mechanisms behind ML concepts
iii. Regression Models
iv. Supervised Models
v. Algorithm Implementation for Supervised Models
vi. Unsupervised Models
vii. Algorithm Implementation for unsupervised Models
viii. Data Preparation for ML models
ix Optimizing Models
x. Feature Engineering with Interpretable models
xi. Sci-kit Learn
xii. Open CV
xiii. Clustering - K Means and K-Medoid
xiv. Bagging and Boosting
xv Time Series Analysis
xvi. Natural Language Processing
xvii. Image Processing
xviii. Neural Networks
xix. Deep Learning Models

Final Assessment - Capstone Project

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